Topic: Linear equations

Question: Find the solution to the linear differential equation.

$$\frac{dy}{dx} + 8x^3y = 12x^3$$

Answer choices:

A $y = Ce^{-2x^4}$

B $y = \frac{1}{2} + Ce^{-2x^4}$

C $y = \frac{3}{2} + Ce^{-2x^4}$

D $y = \frac{5}{2} + Ce^{-2x^4}$

© Krista King Math

Solution: C

The linear differential equation is already in standard form, so we can identify $P(x)$ and $Q(x)$.

$$P(x) = 8x^3$$

$$Q(x) = 12x^3$$

We use $P(x)$ to find the integrating factor.

$$\mu(x) = e^{\int P(x)\ dx}$$

$$\mu(x) = e^{\int 8x^3\ dx}$$

$$\mu(x) = e^{2x^4}$$

Multiply through the differential equation by the integrating factor.

$$e^{2x^4}\left(\frac{dy}{dx} + 8x^3y = 12x^3\right)$$

$$e^{2x^4}\frac{dy}{dx} + e^{2x^4}8x^3y = 12x^3e^{2x^4}$$

Reverse the product rule for derivatives to rewrite the left side,

$$\frac{d}{dx}(e^{2x^4}y) = 12x^3e^{2x^4}$$

then integrate, using a simple substitution with $u = 2x^4$ and $du/8 = x^3\ dx$ to integrate the right side.

$$\int \frac{d}{dx}(e^{2x^4}y) = \int 12x^3e^{2x^4}\ dx$$

© Krista King Math

$$e^{2x^4}y = 12\int e^u\left(\frac{du}{8}\right)$$

$$e^{2x^4}y = \frac{3}{2}e^u + C$$

$$e^{2x^4}y = \frac{3}{2}e^{2x^4} + C$$

$$y = \frac{3}{2} + Ce^{-2x^4}$$

© Krista King Math

Topic: Linear equations

Question: Find the solution to the linear differential equation.

$$xy' + x^2 - x^2y = 0$$

Answer choices:

A $y = 1 + Ce^{\frac{x^2}{2}}$

B $y = 3 + Ce^{\frac{x^2}{2}}$

C $y = 3 + Ce^{\frac{x^2}{4}}$

D $y = 1 + Ce^{\frac{x^2}{4}}$

© Krista King Math

Solution: A

Put the linear differential equation in standard form.

$$xy' + x^2 - x^2y = 0$$

$$y' + x - xy = 0$$

$$y' - xy = -x$$

Now we can identify $P(x)$ and $Q(x)$.

$$P(x) = -x$$

$$Q(x) = -x$$

We use $P(x)$ to find the integrating factor.

$$\mu(x) = e^{\int P(x)\,dx}$$

$$\mu(x) = e^{\int -x\,dx}$$

$$\mu(x) = e^{-\frac{x^2}{2}}$$

Multiply through the differential equation by the integrating factor.

$$e^{-\frac{x^2}{2}}(y' - xy = -x)$$

$$e^{-\frac{x^2}{2}}y' - e^{-\frac{x^2}{2}}xy = -xe^{-\frac{x^2}{2}}$$

Reverse the product rule for derivatives to rewrite the left side,

$$\frac{d}{dx}(e^{-\frac{x^2}{2}}y) = -xe^{-\frac{x^2}{2}}$$

© Krista King Math

then integrate, using a simple substitution with $u = -x^2/2$ and $du = -x\ dx$ to integrate the right side.

$$\int \frac{d}{dx}(e^{-\frac{x^2}{2}}y) = \int -xe^{-\frac{x^2}{2}}\ dx$$

$$e^{-\frac{x^2}{2}}y = \int e^u\ du$$

$$e^{-\frac{x^2}{2}}y = e^u + C$$

$$e^{-\frac{x^2}{2}}y = e^{-\frac{x^2}{2}} + C$$

Solve for y to find the solution to the linear differential equation.

$$y = 1 + Ce^{\frac{x^2}{2}}$$

© Krista King Math

Topic: Linear equations

Question: Find the solution to the linear differential equation.

$$xy' - 2y = x^2$$

Answer choices:

A $y = x^2 \ln x + C$

B $y = \dfrac{x^2}{\ln x}$

C $y = Cx^2 \ln x$

D $y = x^2(\ln x + C)$

© Krista King Math

Solution: D

Put the linear differential equation in standard form.

$$xy' - 2y = x^2$$

$$y' - \frac{2}{x}y = x$$

Now we can identify $P(x)$ and $Q(x)$.

$$P(x) = -\frac{2}{x}$$

$$Q(x) = x$$

We use $P(x)$ to find the integrating factor.

$$\mu(x) = e^{\int -\frac{2}{x}\,dx}$$

$$\mu(x) = e^{-2\ln x}$$

$$\mu(x) = e^{\ln x^{-2}}$$

$$\mu(x) = x^{-2}$$

Multiply through the differential equation by the integrating factor.

$$x^{-2}\left(y' - \frac{2}{x}y = x\right)$$

$$x^{-2}y' - 2x^{-3}y = x^{-1}$$

Reverse the product rule for derivatives to rewrite the left side,

© Krista King Math

$$\frac{d}{dx}(x^{-2}y) = x^{-1}$$

$$\frac{d}{dx}\left(\frac{y}{x^2}\right) = \frac{1}{x}$$

then integrate.

$$\int \frac{d}{dx}\left(\frac{y}{x^2}\right) = \int \frac{1}{x}\,dx$$

$$\frac{y}{x^2} = \ln x + C$$

Solve for y to find the solution to the linear differential equation.

$$y = x^2(\ln x + C)$$

© Krista King Math